A Mai Ly



Senior full-stack & backend engineer

I build products end to end for European teams, with most of my depth in the backend: APIs, realtime systems and AI features that stay up in production.

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About

Since 2020 I've been an engineer at Digital Unicorn, a French software studio. I've worked on more than 20 of its products, usually as the main backend engineer: NestJS and Node APIs, payment flows, realtime features, and LLM and voice integrations, deployed on AWS and Azure with Docker and Kubernetes.

I build the front end too: React and Next.js on the web, Flutter on mobile, so one person can own a feature from database to UI. Most of my work is with French clients in cross-border teams, so I write specs and reviews in English, mentor newer developers and take ownership of what happens in production.

Skills

Core stack	**TypeScript, Node.js, NestJS, React, Next.js, Flutter, PostgreSQL, MongoDB, Redis, GraphQL, Docker, Kubernetes, AWS**
Backend	NestJS, Node.js, Express, AdonisJS, TypeScript. REST, GraphQL (Apollo, Hasura), gRPC, WebSockets, SSE, Socket.IO. Microservices, event-driven design, queues (BullMQ, Kafka), CQRS and DDD, OAuth2, JWT, RBAC, webhooks, rate limiting, caching, observability.
Front end and mobile	React, Next.js 14 and 15, Vue and Nuxt, React Native, Flutter, Tailwind CSS, shadcn/ui, Ant Design.
AI and voice	OpenAI (GPT-4, 4o, 5; Assistants), Azure OpenAI, LangChain, ChromaDB and RAG, Retell AI, Deepgram, ElevenLabs, DeepL, prompt design.
Data	PostgreSQL, MongoDB, MySQL, Redis, Elasticsearch, Supabase. Prisma, TypeORM, Sequelize, Mongoose.
Cloud and DevOps	AWS (EC2, ECS, EKS, Lambda, S3, SES, CloudFront, Route 53), Azure (AKS, Pipelines), Cloudflare, Docker, Kubernetes, Helm, ArgoCD, GitLab CI, Nginx, PM2.
Integrations	Stripe, MangoPay, RevenueCat, Google APIs (Auth, FCM, Calendar), Firebase, Algolia, Twilio, Brevo, Web3.js, ethers.js, Thirdweb.

Education & certificates

Top Notch 2 English certificate, CEFR level B2

FPT Polytechnic, Software applications, 2016 - 2018. GPA 7.8/10.

Languages

Vietnamese	Native
English	Professional working proficiency (B2)

Experience

2024 - now

Senior full-stack engineer

Digital Unicorn Co., Ltd, full-time

- Lead engineer on Nutrissimo, a realtime multiplayer game for Institut Pasteur de Lille that launches inside Moodle through LTI 1.3.
- Backend lead on NightPass ticketing. Fixed a Redis out-of-memory incident and made rolling deploys safe during live events.
- Built AI products end to end: Days Legacy, RP Tech AI, SparkPlan with its voice agents, and CLS AI.
- Own architecture, delivery and production reliability, and I'm often the top contributor to the codebase.

2020 - 2024

Backend and full-stack engineer

Digital Unicorn Co., Ltd, full-time

- Main backend engineer on WeShake (Stripe and MangoPay payments) and OOVOOM (ran the AWS EKS and Helm deployments).
- Lead backend engineer on BeParentalis and RP Tech AI.
- Also built DeliverLIB, Just Meat, Raisy, Kibo and CyLimit: payments, realtime features, token vesting and role-based access.

2019 - 2020

Software developer

Phuc Quan Service Trading Co., Ltd

Designed relational databases and built Node.js REST APIs. Researched tools, tuned performance, and debugged and refactored existing modules.

2019

Software developer

G-Lance

My first job. Built Nail Schedule, a booking product (June to December 2019): Node.js REST APIs and the database schema.

Selected work

Nutrissimo

2026 - now

Lead full-stack engineer, for Institut Pasteur de Lille

- Built across the whole Turborepo: the Flutter app, the Next.js admin and Supabase.
- Built the server-authoritative game engine: presence, reconnection after a dropped connection, and idempotent state updates.
- Implemented LTI 1.3 single sign-on so the game launches straight from Moodle.

Flutter, Next.js 15, Supabase Realtime, Turborepo, LTI 1.3, Cloudflare

NightPass

nightpass.fr

Backend lead, ticketing for venues and events. About 10k users on web and mobile.

- Traced a production out-of-memory incident to a Redis reconnect storm; fixed the config and added leader locks so rolling deploys don't drop traffic.
- Reworked the QR scan path (back-pressure, compare-and-set writes, readiness probes) to stay fast with many doors scanning at once.

NestJS, GraphQL, Prisma, PostgreSQL, Redis, BullMQ, Stripe, AWS ECS

SparkPlan

sparkplan.ai sparkplanvoice.com

AI engineer, budget advisor running on Sortlist.com. About 5k reports generated.

- Wrote most of the integration backend.
- Built the OpenAI, LangChain and ChromaDB pipeline: prompts, routing between GPT-4o, GPT-4 and GPT-3.5, and answers in the user's language across five EU locales.

NestJS, OpenAI, Azure OpenAI, LangChain, ChromaDB, WebSocket, MongoDB, Stripe

CLS AI for Promecano

AI and backend engineer, voice assistant for industrial IoT diagnostics.

- Connected Retell AI voice to OpenAI Assistants with one vector store per device.
- Streamed tool status to the client over WebSockets and moved slow work to Bull queues with caching to keep latency low.

NestJS, OpenAI, Retell AI, vector search, WebSockets, MongoDB, Kubernetes

RP Tech AI

Backend architect, AI productivity suite for French users. About 8k users.

- Built the backend almost alone.
- Combined four providers in one service: OpenAI, Rev for speech-to-text, Ordalie for legal answers and DeepL for translation, with LangChain and ChromaDB for retrieval.

NestJS, OpenAI, Rev, DeepL, Ordalie, LangChain, ChromaDB, MongoDB, Stripe

BeParentalis

Lead backend engineer, parenting app reviewed by pediatricians. About 20k downloads, rated 4.6.

- Designed the data model linking doctors, families, lessons and subscriptions.
- Integrated Algolia search and RevenueCat for in-app subscriptions on both stores.

NestJS, MySQL, TypeORM, Algolia, RevenueCat, Firebase, Brevo

WeShake

weshake.io

Senior backend engineer, peer-to-peer payments and cards. About 12k users.

- Integrated two payment processors, Stripe and MangoPay, with payment links, recurring billing and webhooks.
- Shipped GDPR data deletion, multi-device login with OTP, and the referral and bonus system.

Node.js, Express, MySQL, Sequelize, Stripe, MangoPay, Socket.IO, Twilio, AWS, Kubernetes

OOVOOM

oovoom.com

Backend developer and DevOps, ERP for a garage network. 100+ partner garages, about 6k users.

- Designed role-based access for owners, garage staff, network staff and admins.
- Automated quotes and invoices, and ran the AWS EKS and Helm deployment pipeline.

Node.js, Express, MySQL, Sequelize, Elasticsearch, Algolia, AWS EKS, Helm, Stripe

Other projects

Days Legacy (DSL)
AI advisor for wealth and legacy planning. It combines a user's portfolio with live financial news to answer planning and risk questions. Full-stack, 2024.

CyLimit
Fantasy cycling with rider NFTs that earn from real race results. I built the server-sent announcements, notifications and media pipeline.

Eddwin
Marketplace for training courses. I built the multilingual Next.js front end, onboarding for learners and trainers, Bull email automation and Stripe billing for five trainer tiers.

Profiler Connection and Raisy
Matchmaking app plus a Web3 deal platform, about 5k users. I built the token vesting engine (TGE, cliffs, linear release) and investor cap tables.

DeliverLIB
Food delivery marketplace with shops, catalogues, live order tracking and dynamic delivery pricing.

Just Meat
Online butcher with nested categories, courier assignment and scheduled delivery slots.

Venues
Venue management with Elasticsearch search, bookings that handle time zones, and an ArgoCD deployment on Kubernetes.

Kibo Finance
Loyalty backend for a Web3 app: points, leaderboards, referral codes and on-chain checks.

ICF Crawler
Crawler that collects listings from 100+ luxury fashion stores, handling bot protection and reCAPTCHA.